

SELECTIVE COORDINATION STUDY — TCC1

Project: synthetic example · Bus: B-12.47kV · Drawing: E-501 Rev A

Log-log time-current characteristic curves below cover the primary fuse, secondary main fuse, 200 A feeder fuse, and 20 A branch breaker. Coordination is verified at the system fault duty.

#	Device	Type	Rating	Curve	TD
①	Primary Fuse	Class L	1600 A	—	—
②	Sec Main Fuse	Class K-5	400 A	—	—
③	Feeder Fuse	Class RK1	200 A	—	—
④	Branch Breaker	Thermal-magnetic	20 A	Inverse	—
⑤	Conductor	#6 THWN-2 Cu	—	—	—
⑥	Motor M-101	Induction 75 HP	104 A FLC	—	—
⑦	MV Relay 51P	GE Multilin 750	600 A	U2 IEC VI	0.55
⑧	MV Relay 51N	GE Multilin 750	75 A	U2 IEC VI	0.40

TCC1 reference points (per device on log-log axes)

Device	Pickup (A)	Time @ Pickup (s)	Asymptote (s)
①	1600	100	0.01
②	400	60	0.05
③	200	30	0.1
④	20	30	0.5

References: IEEE Std 242 (Buff Book) §10 · IEEE C37.91 · Bussmann Coordination Bulletin 201